Supply, Demand, and Government Policies

Start from market equilibrium → impose a price control → analyze a tax → explain who bears it

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1. Begin with the market: what do prices do?

A market price is not just a number on a receipt. It conveys information about how much buyers want and how much sellers are prepared to supply. We already know that, other things equal, a lower price raises quantity demanded and reduces quantity supplied. A higher price has the opposite effects. A competitive market tends toward the price where these quantities are equal.

> **DEFINITION Equilibrium, shortage, and surplus**
>
> **Market equilibrium** is the price at which quantity demanded equals quantity supplied. A **shortage** occurs at a given price when **quantity demanded exceeds quantity supplied**. A **surplus** occurs when **quantity supplied exceeds quantity demanded**.

Suppose that 100 secure bicycle lockers near a commuter station are rented per week at a price of $30 each. At that price, the number people wish to rent equals the number operators wish to make available. Now suppose a rule changes what price can legally be charged. Demand and supply still matter: a legal price does not automatically produce the quantity people want.

> **KEY RESULT The question for every government policy**
>
> Identify the new legal constraint or tax. Then ask: **What price can buyers pay? What price do sellers receive? How many units will buyers demand? How many will sellers supply? How many transactions will actually occur?** These can be different quantities or prices.

PART A Supply and demand: price ceilings and price floors

2. Price ceilings: a legal maximum

> **DEFINITION Price ceiling and binding ceiling**
>
> A **price ceiling** is a legal **maximum** price. A ceiling **above** the unrestricted market equilibrium is **nonbinding**: the market can still trade at its usual price. A ceiling **below** equilibrium is **binding**: it prevents price from rising to the market-clearing level.

Worked example: A limit on the weekly price of bicycle lockers

Imagine a city limits weekly locker rentals to $20. Before the rule, the equilibrium price was $30 and 100 lockers were rented. At the legally permitted $20 price, 140 people would like a locker but operators are willing to supply only 60.

Price per week	Quantity demanded	Quantity supplied	Market outcome
\$20 (binding ceiling)	140	60	Shortage of 80
\$30 (unrestricted equilibrium)	100	100	Market clears
\$36 (nonbinding ceiling)	Market still trades at \$30		No effect

Why is there a shortage? The lower price makes renting more attractive, so more people want lockers. But fewer lockers are profitable to provide. The ceiling prevents the price rising to eliminate the gap. **Only up to 60 lockers are available for rental**, not 140. The shortage of 80 is *unmet demand*, not 80 additional transactions.

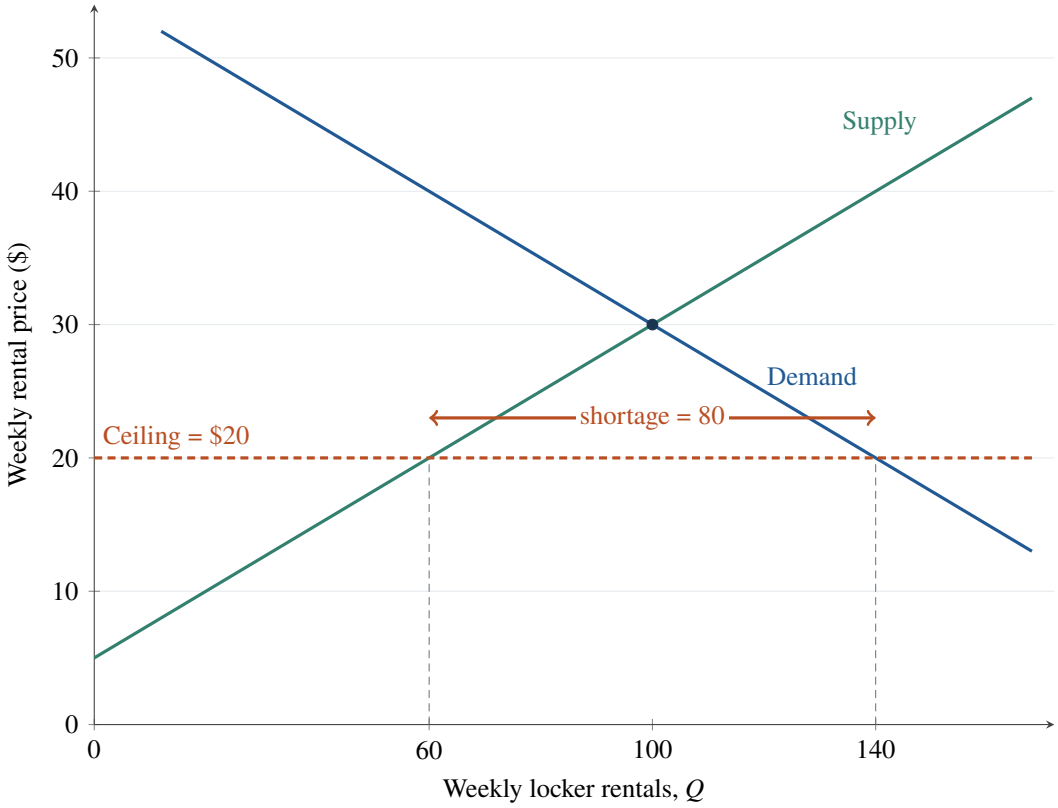


Figure 1. A binding price ceiling causes a shortage. At \$20, suppliers offer 60 lockers, while renters demand 140. The 80-unit gap cannot be removed by a legal price increase.

Read the graph. Start at the horizontal \$20 ceiling. Move to the *supply* curve to find 60 lockers; move to *demand* to find 140. The horizontal distance between them is the shortage. The original equilibrium (100 lockers at \$30) is no longer allowed. A ceiling of \$36 would instead be nonbinding: firms could still charge the equilibrium price of \$30.

KEY RESULT What a binding price ceiling actually does

Price below equilibrium ⇒ shortage: $Q_d > Q_s$**.** In the basic competitive model, the actual quantity traded is limited by the smaller quantity, Q_s. Some customers who obtain the good pay less, but others cannot obtain it at all. The lower posted price does *not* guarantee improved access for every buyer.

3. Why shortages have costs beyond the posted price

When more people want a good than there are units available, someone must decide who receives it. The price can no longer perform all the rationing. Waiting lists, time spent searching, personal connections, and allocation rules may replace it. These mechanisms can waste time or allocate units to people who do not value them most highly.

Short run versus long run. In the short run, existing housing or other facilities cannot change quickly. Over time, suppliers can reduce investment or maintenance and buyers can alter their choices. If supply and demand become more responsive to price, a continuing binding ceiling can produce a larger shortage. This is one reason the consequences of rent controls cannot be judged solely from the first month after a restriction.

A shortage is not proof that *every* consumer is worse off. A person who actually rents a controlled unit at a lower price may benefit. The economic question is who obtains the scarce units, how many units remain available, and what other costs the rules create.

4. Price floors: a legal minimum

> **DEFINITION Price floor and binding floor**
>
> A **price floor** is a legal **minimum** price. A floor **below** unrestricted equilibrium is **nonbinding**; a floor **above** equilibrium is **binding**. A binding floor leads to **quantity supplied exceeding quantity demanded**, in the basic competitive model.

Worked example: A minimum rental price for the same lockers

Now suppose a different hypothetical rule requires operators to charge *at least* $40 per locker. At $40, operators would supply 140 lockers, while renters want only 60. The result is a surplus of **80 available but unwanted lockers**. Only 60 can be rented at that price, unless someone (such as the government) buys the unwanted supply. A $22 minimum, by contrast, would be nonbinding because the market already charges $30.

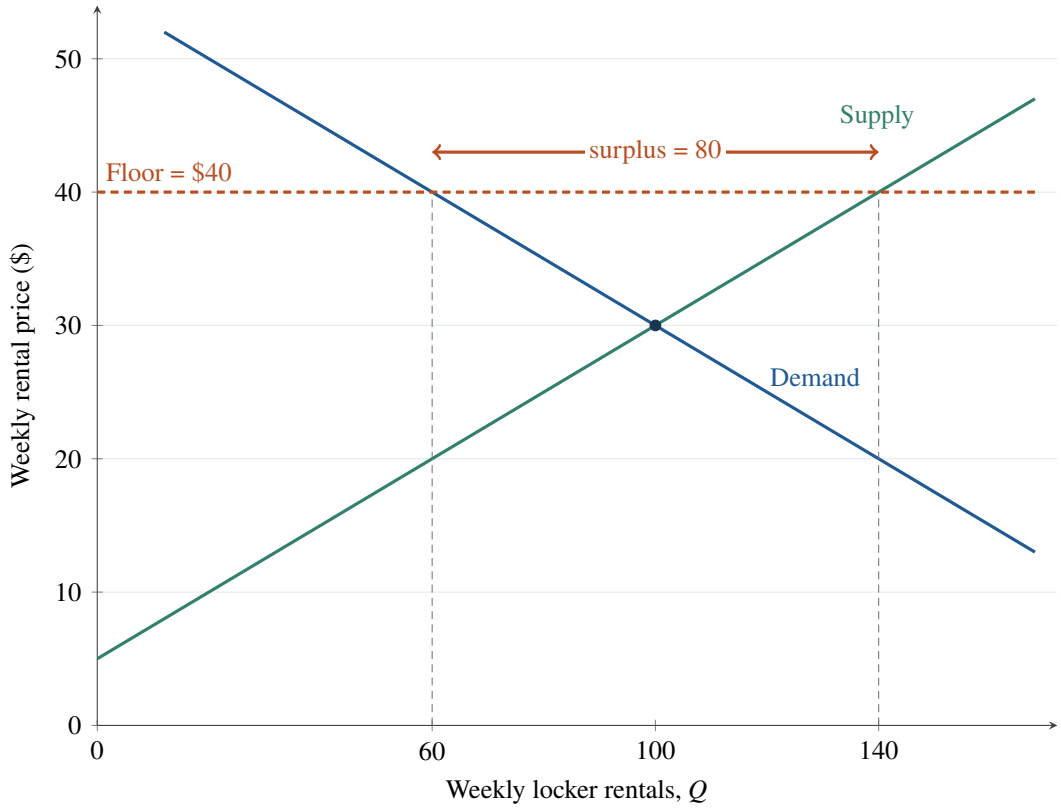


Figure 2. A binding price floor creates a surplus. At the legal minimum of $40, renters demand 60 lockers, but suppliers would make 140 available.

Read the graph. Follow the $40 horizontal floor to the demand curve to find 60 and to supply to find 140. The 80-unit surplus is the difference. There is no shortage: sellers wish to sell *more* than customers wish to buy. Nor do 140 lockers actually get rented just because operators would like to supply them.

> **KEY RESULT Ceilings and floors are opposites**
>
> **Binding ceiling below equilibrium: shortage, $Q_d > Q_s$.**
> **Binding floor above equilibrium: surplus, $Q_s > Q_d$.**
> A nonbinding control does not alter the market equilibrium. At a price exactly equal to the original equilibrium, the control leaves that equilibrium feasible and has no immediate effect in the simple model.

5. The minimum wage is a price floor in a labor market

A wage is the price of an hour of labor. Workers *supply* labor; employers *demand* labor. Thus, the supply and demand roles are reversed from a market in which households buy a finished good. A minimum wage sets the lowest legal wage for covered jobs.

Worked example: Hourly event-staff work in a competitive market

Suppose, in a hypothetical competitive market, the equilibrium wage is $18 per hour and 80 hours of work are hired per week. If a $22 minimum wage is imposed, employers wish to hire 60 hours while workers offer 100 hours. The gap is **40 hours of excess labor supplied**. Employment in this simple example is determined by the 60 hours employers hire, not by the 100 hours workers would like to sell.

Hourly wage	Labor hours demanded	Labor hours supplied	Gap
$18 (equilibrium)	80	80	0
$22 (minimum wage)	60	100	40 excess hours

This does *not* mean exactly 40 people become unemployed: the quantities are hours, not individuals. Workers who keep their jobs may earn a higher hourly wage; other workers may find fewer hours available. A minimum below $18 would not bind in this particular market.

> **KEY RESULT What the competitive model predicts, and what it does not**
>
> For a **binding minimum wage in a competitive labor market**, the higher wage reduces employers' quantity of labor demanded and raises the quantity workers wish to supply. **Excess labor supply** results. This is a model prediction, not a universal numerical estimate of actual job losses. Real outcomes depend on market conditions, coverage, enforcement, and whether employers have wage-setting power.

6. Why governments use price controls, and other options

A ceiling can aim to make an essential good affordable; a floor can aim to raise a seller's or worker's earnings. These are distributional objectives, not claims that the market's supply-and-demand curves disappear. The difficulty is that controlling the price also changes incentives to buy, sell, invest, and search.

Alternatives. A targeted housing transfer can help qualifying renters pay market prices rather than impose one low rent on all units. A wage subsidy or an earned-income tax credit can raise eligible workers' take-home income without directly making it illegal for firms to hire at the original market wage. These policies can still affect demand or supply and must be financed; their effects are not costless. Which policy is preferable depends on the objective and its full costs.

PART B Taxes: the tax wedge and elasticity

7. A tax creates two prices, not one

A government may impose a *per-unit tax* on a transaction to raise revenue. For example, it might collect $6 for every bicycle tune-up sold. The effect is not simply that a business hands over $6. Buyers and sellers respond to the tax, and the market price changes.

DEFINITION Per-unit tax, tax wedge, and tax incidence

A **per-unit tax** is a fixed amount paid to government for each unit traded. Let P_B be the total price paid by the buyer, P_S the amount the seller keeps after the tax, and t the tax per unit. The **tax wedge** is

$$\boxed{P_B - P_S = t.}$$

Tax incidence means **how the economic burden is divided between buyers and sellers**, as opposed to who is legally required to send the tax to government.

Worked example: A $6 tax on bicycle tune-up appointments

Before the tax, 160 tune-ups are sold each week at $26 apiece. After a $6 tax, suppose the market settles at 120 appointments: buyers pay $30 per appointment, while mechanics keep $24 after the tax. The $6 difference is sent to government.

Outcome	Before tax	After tax
Weekly appointments traded	160	120
Price paid by buyers	$26	$30
Price received by sellers, net of tax	$26	$24
Difference sent to government per appointment	$0	$6

Who bears the $6? Buyers now pay $4 more ($30 instead of $26). Sellers retain $2 less ($24 instead of $26). So the $6 burden is split: **$4 on buyers and $2 on sellers**. The government receives $120 \times \$6 = \$720$ per week, not $160 \times \$6$. The number of sales has fallen.

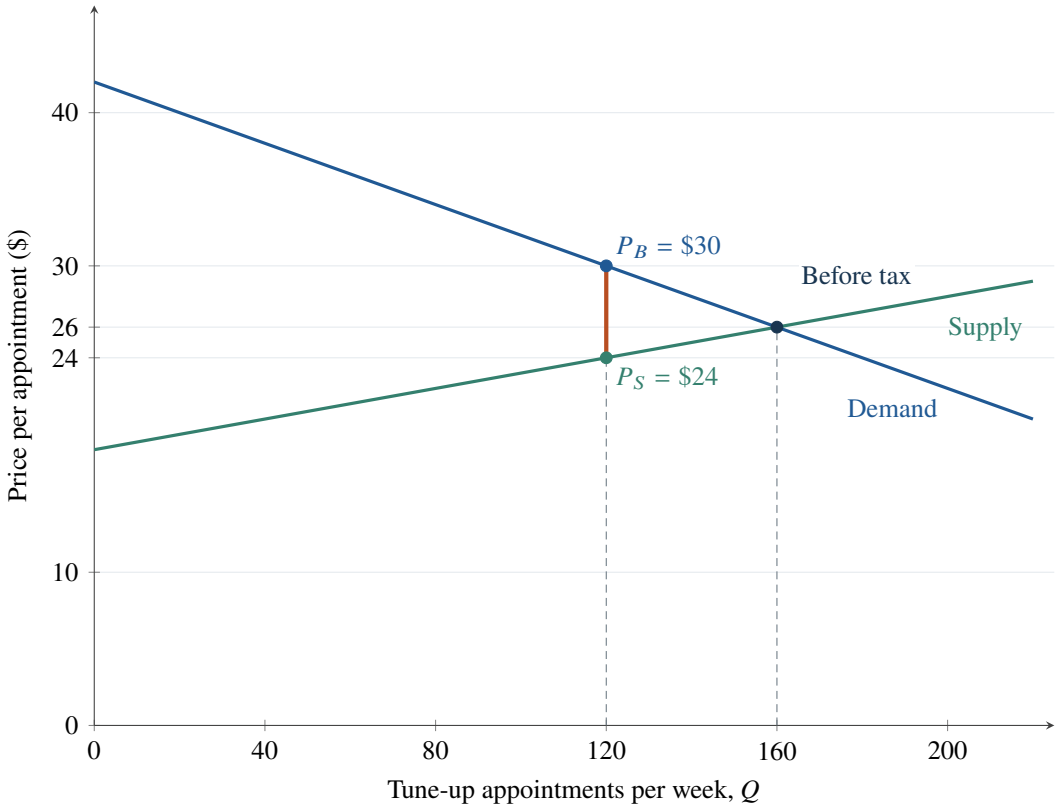


Figure 3. The tax wedge. At 120 appointments, buyers are willing to pay $30, sellers require $24, and the difference is the $6 tax. Before tax, 160 appointments traded at $26.

Read the graph. At $Q = 120$, move up to demand to find the buyer's $30 price, and to supply to find the seller's $24 net price. The orange vertical segment is the $6 tax. At the old quantity of 160, the two curves met at $26; those extra 40 appointments no longer take place. A *tax wedge* connects the new buyer and seller prices at the same traded quantity.

> **KEY RESULT Three effects of a per-unit tax in the standard competitive model**
>
> **1.** The quantity traded normally falls. **2.** Buyers pay a higher total price and sellers keep a lower net price (when both sides respond). **3.** Government tax revenue equals **tax per unit × quantity actually traded**. Tax receipts are transfers to government, not additional goods or services produced.

8. Does it matter whether the law taxes buyers or sellers?

> **KEY RESULT Legal payment is not the same as economic incidence**
>
> In the standard competitive model, a **$6 tax on buyers** and a **$6 tax on sellers** create the same $6 wedge between what buyers pay and what sellers keep. With the same demand and supply, the resulting traded quantity and the **economic split of the burden are the same**. The legal responsibility for sending money to the government does not determine who ultimately bears it.

If sellers submit the tax: the buyer hands $30 to the mechanic. The mechanic sends $6 to government and keeps $24. The tax raises the seller's required *gross* selling price at every quantity: draw the supply curve $6 higher when prices are measured as amounts buyers pay.

If buyers submit the tax: the buyer pays the mechanic $24 and separately sends $6 to government. The mechanic keeps $24. The tax reduces the amount buyers can offer sellers at a given quantity: draw the demand curve $6 lower when prices are measured as amounts sellers receive.

In either case, **the buyer's full outlay is $30; the seller's net receipt is $24; 120 appointments are traded**. This result assumes the same market and tax rules, with competitive price adjustment. The words "tax on sellers" do not mean sellers necessarily absorb the whole tax.

9. Elasticity determines who bears more of the tax

The important question is not who mails the payment. It is *which side can respond more easily to an unfavorable change in price*. If customers can readily stop buying, sellers must absorb more of a tax to keep customers. If sellers can readily stop supplying, buyers must shoulder more of it to keep the product available.

DEFINITION Price elasticity of demand and supply

Price elasticity measures how strongly quantity responds to price. In words,

$$\text{Price elasticity} = \frac{\text{percentage change in quantity}}{\text{percentage change in price}}.$$

Demand elasticity is usually expressed as an *absolute value*. **Inelastic** means the percentage change in quantity is smaller in magnitude than the percentage change in price; **elastic** means it is larger. You do not need calculus to use these ideas.

KEY RESULT The main tax-incidence rule

The less price-elastic side of the market bears more of the economic burden of a tax.
If demand is relatively inelastic and supply relatively elastic, **buyers bear more**.
If supply is relatively inelastic and demand relatively elastic, **sellers bear more**.
When both respond similarly, the burden is more evenly divided.

Worked example: Same $10 tax, different responsiveness

Consider two *separate hypothetical markets*. Both originally trade 100 units at $30. Both then face a $10 tax, and both trade 50 units after the tax. What differs is the way buyers and sellers respond to prices.

Market	Buyer price	Seller receipt	Buyer burden	Seller burden
Demand less elastic than supply	$37.50	$27.50	$7.50	$2.50
Supply less elastic than demand	$32.50	$22.50	$2.50	$7.50

Each row checks: the buyer and seller prices differ by exactly $10, and the buyer and seller burdens add to $10. The next two graphs show *why* the split differs.

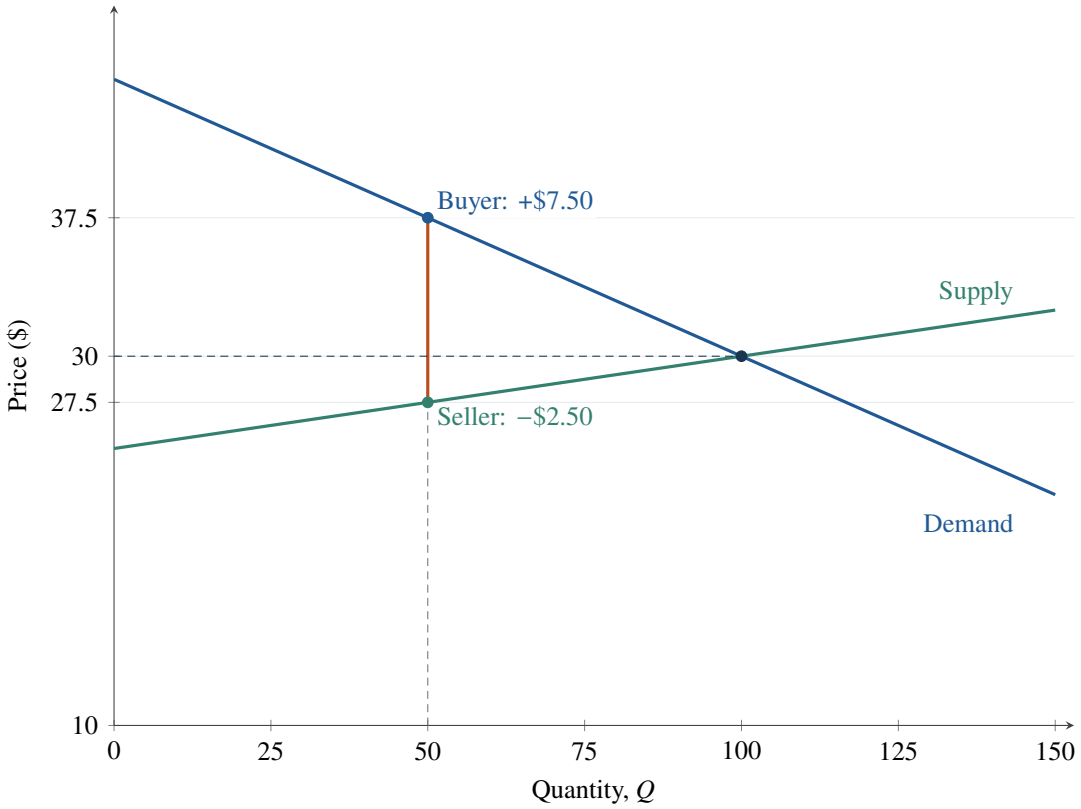


Figure 4. Demand is less elastic than supply. Buyers have fewer easy alternatives, so their price rises by $7.50 of the $10 tax; sellers' net price falls by $2.50.

Read the graph. Compare both new prices with the original $30 line. The buyer's price is substantially *above* it; the seller's price is only slightly *below*. The more inelastic buyers absorb most of the tax. Notice that the $10 tax is *not* a $10 increase in the buyer's price.

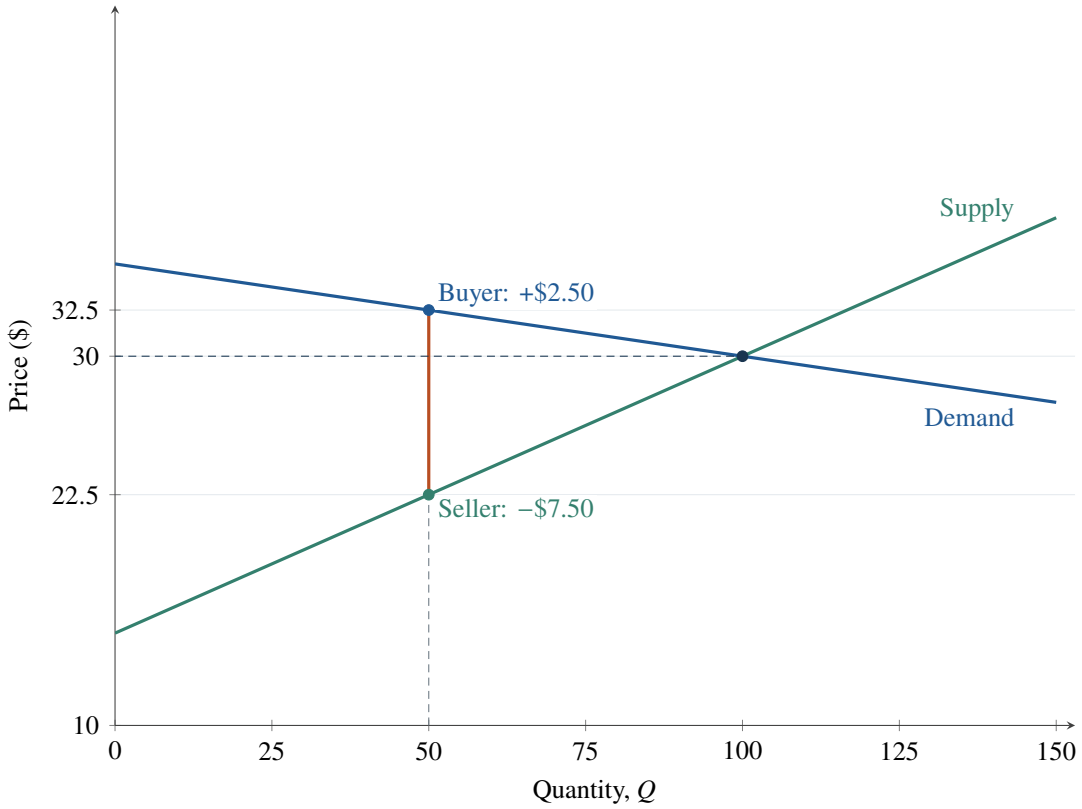


Figure 5. Supply is less elastic than demand. Sellers have fewer easy alternatives, so their net price falls by $7.50 of the $10 tax; buyers' price rises by only $2.50.

Read the graph. The original price is again $30, and the tax is again $10. Buyers can cut purchases relatively easily, so the price they pay rises only to $32.50. Sellers cannot withdraw supply as easily, and their net price falls to $22.50. **The same statutory tax creates a different division of the burden because the relative elasticities differ.**

> **KEY RESULT Why the rule works: economic reasoning, not a slogan**
>
> When demand is *inelastic*, customers keep buying even after a price increase, so sellers can pass on more of the tax. When supply is *inelastic*, sellers cannot easily move their labor, equipment, or goods elsewhere, so they accept a larger fall in their net selling price. **The side with fewer alternatives has less ability to avoid the tax.**

Method for a new question. For a price control, identify the legal limit, decide whether it binds, then compare Q_d and Q_s *at that price*. For a tax, compare the new buyer and seller prices with the original price and check that their difference equals the tax; revenue is tax times *actual* sales. For incidence, find which side is less elastic. These predictions use the standard competitive model; market power and other complications require further analysis.